

GEOS-MPM Examples Suite Report

Generated by examples/runExampleSuite

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Running and rerunning

Each example directory contains a copyable `pfw_input_*.py` and a `runProblem` wrapper. The wrapper stages an isolated case directory under `testRunDirectory`, copies the needed PFW files, runs the particle-file writer, submits GEOS, submits dependent post-processing, and copies figures/logs back to the example `output/` directory.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/examples
./runExampleSuite all --python /usr/tce/bin/python3 --force
./runExampleSuite all --case borehole_Ghareb --python /usr/tce/bin/python3 --force
./runExampleSuite report --python /usr/tce/bin/python3
```

If output exists, `runProblem` prompts before overwriting. Use `--force` for noninteractive reruns.

Timing summary

Example	Family	GEOS job	GEOS state	GEOS wall	GEOS CPU	Post wall	Post CPU
borehole_Ghareb	Borehole collapse	6044177	COMPLETED	00:00:03	00:00:00	00:00:01	00:00:00

borehole_Ghareb

A plane-strain borehole collapse demonstration using the Ghareb geomechanics parameterization from `pfw_materials.py`. The pressure events initialize confinement and unload the borehole wall.

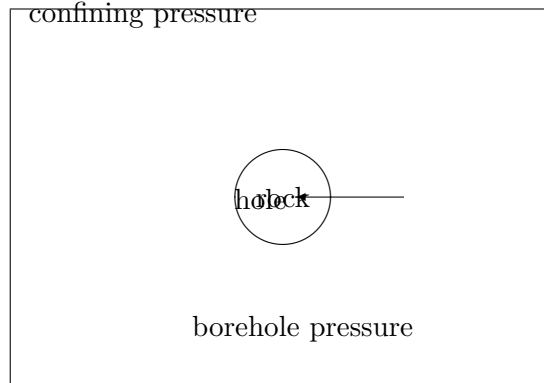


Figure 1: Schematic for borehole_Ghareb.

Code features exercised.

- Geomechanics material model
- borehole pressure MPM event
- confining pressure event
- Voronoi-Weibull flow field
- plane strain CPDI

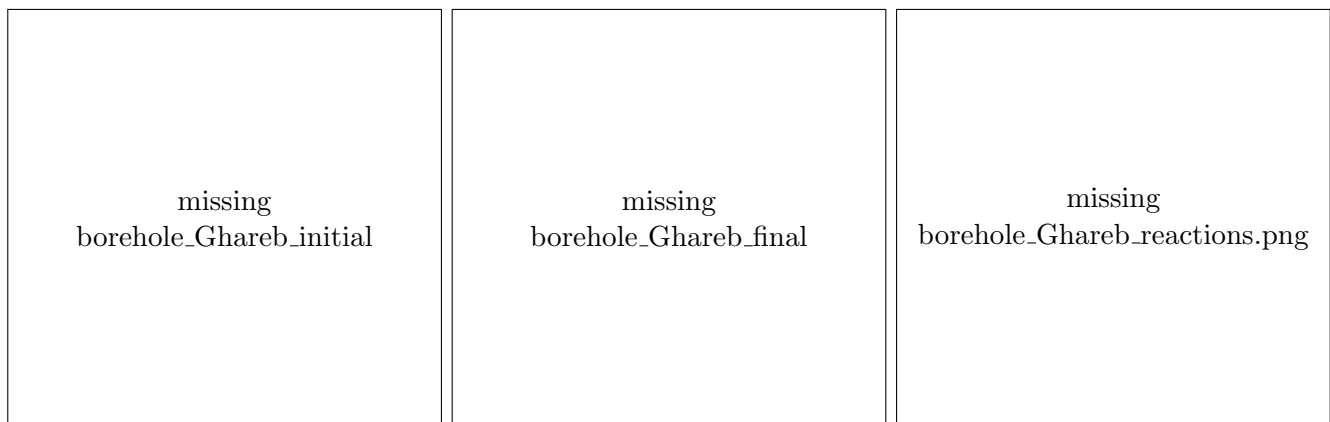


Figure 2: Initial state, final state, and reaction history for borehole_Ghareb.